
Non-Convex Distributed Optimization without Full Client Participation under Second-Order Similarity

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We study non-convex distributed optimization under the second-order data sim-
2 ilarity assumption. Existing homogeneity-aware methods achieve optimal com-
3 munication complexity in terms of client–server vector exchanges. However, they
4 rely on periodic rounds of full client participation. This requirement constitutes
5 a significant drawback, increasing the training time. In contrast, algorithms that
6 avoid full participation fail to exploit the homogeneity of local datasets to accelerate
7 convergence. In our work, we propose NFG-SS, a novel distributed algorithm that
8 leverages second-order similarity while completely avoiding full client participa-
9 tion. Our method is motivated by a principled design choice and supported by
10 a rigorous convergence analysis. Notably, NFG-SS does not require convexity of
11 the objective, which is uncommon for this class of algorithms. Empirical results
12 demonstrate the superiority of our approach over both homogeneity-aware methods
13 and schemes without full participation. Overall, our work bridges the gap between
14 these two key paradigms in distributed learning.

15 1 Introduction

16 The rapid growth of modern machine learning models and datasets has made large-scale optimization
17 a central computational bottleneck [Bottou et al., 2018]. Distributed learning addresses this challenge
18 by allowing multiple nodes to perform local computations while a central server coordinates model
19 updates [Verbraeken et al., 2020]. In the standard client-server setting, the server maintains the global
20 model, whereas clients compute local update directions using their private datasets. Formally, we
21 consider the finite-sum optimization problem

$$\min_{x \in \mathbb{R}^d} \left[f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x) \right] \quad \text{with} \quad f_i(x) = \frac{1}{m_i} \sum_{j=1}^{m_i} \ell(g(x, a_j^i), b_j^i) \quad (1)$$

22 where n is the number of nodes, m_i is the size of the local dataset at node i , $g(x, a_j^i)$ denotes the
23 model prediction with trainable parameters x , and (a_j^i, b_j^i) is the j -th data point stored at node i . For
24 algorithmic purposes, we designate f_1 as the server-side objective and treat $f_2, \dots, f_n$ as client-side
25 objectives. This convention is used throughout the paper because our method performs proximal
26 updates with respect to f_1 .

27 In distributed learning, transmitting model updates is typically far more expensive than performing
28 local computation on each device [Konečný et al., 2016, Kairouz and McMahan, 2021]. Therefore,
29 the main challenge is to design methods that achieve superior performance while keeping the
30 communication cost low. One prominent direction in round-efficient optimization is the exploitation
31 of local steps, where each client performs multiple local updates between communication rounds,
32 as in FedAvg and FedProx [McMahan et al., 2017, Khaled et al., 2020, Woodworth et al., 2020, Li
33 et al., 2020].

In a distributed setting, the data on different clients comes from a common underlying distribution, since a single training dataset is split across workers Zinkevich et al. [2010], Dekel et al. [2012]. In this setup, the local objectives of clients are not identical. The degree of homogeneity between them is the key factor that determines the sustainability of local steps Karimireddy et al. [2020], Woodworth et al. [2020]. When local objectives are more similar, the local loss landscape of each client more faithfully reflects the global landscape, and local updates push iterates in directions consistent with the global function. Following this intuition, recent studies formalize homogeneity through the second-order similarity assumption [Kovalev et al., 2022, Lin et al., 2023, Karagulyan et al., 2024], also known as the δ -relatedness assumption or bounded Hessian dissimilarity. Exploiting this assumption helps distributed methods accelerate convergence, since in practice $\delta \ll L$ Arjevani and Shamir [2015], Kovalev et al. [2022].

Definition 1 (Second-order similarity). *We say that the local objectives $\{f_i\}_{i=1}^n$ satisfy the δ -second-order similarity condition if for all $x \in \mathbb{R}^d$ and all $i \in [n]$*

$$\|\nabla^2 f_i(x) - \nabla^2 f(x)\| \leq \delta. \quad (2)$$

In particular, if local objectives are constructed from i.i.d. samples, learning theory implies that δ decreases as $\mathcal{O}(1/\sqrt{m})$, where m is the local dataset size [Hendrikx et al., 2020]. When δ is small, each client’s information becomes more relevant for longer periods of time, meaning that every worker can do more local steps. Remarkably, algorithms such as FedAvg and FedProx do not exploit this regime and their convergence rate does not improve as the homogeneity between clients grows, so the benefits of similarity remain unused.

Early communication-efficient methods aimed to minimize number of communication rounds [McMahan et al., 2017, Stich, 2018, Kovalev et al., 2022]. This metric is appropriate when devices and communication channels are considered to be equivalent. However, it is not the case of real-world systems. The server cannot aggregate updates from all clients simultaneously, and even a single full synchronization round transmits $\mathcal{O}(n)$ vectors. Therefore, one should also consider the total amount of transmitted vectors, which counts cost per round and their number. This naturally motivates partial participation, where only a subset of clients operates with server at a time. McMahan et al. [2017]. Under this measure, existing distributed methods with theoretical guarantees for local steps still require full aggregation across all clients per round [Stich, 2018, Khaled et al., 2020], which alone incurs $\mathcal{O}(n)$ transmitted vectors per round. The same limitation appears in similarity-based methods, where periodic full-gradient refreshes implicitly require synchronized participation of all clients [Kovalev et al., 2022, Lin et al., 2023].

On the other hand, algorithms that eliminate full gradient calculation are developed for centralized problems, and do not consider local updates Li et al. [2021b], Medyakov et al. [2025]. If one transfers these methods to the distributed setting, where each gradient evaluation ∇f_i corresponds to communication with a client, the convergence rate will depend on smoothness constant L , rather than the δ . Thus, making them insensitive to data homogeneity across clients. This leaves open the question of whether one can exploit second-order similarity in distributed non-convex optimization and simultaneously avoid explicit full-gradient computation with theoretical guarantees.

Our Contribution. We design a distributed method with local steps. Unlike existing similarity-based distributed methods, **our approach never computes or aggregates the full gradient at any step**. Second-order similarity is crucial in this setting, since it ensures that information obtained from sampled clients remains relevant to the global objective. To the best of our knowledge, this combination of complete no-full-gradient computation paradigm and second-order similarity is not covered by existing works.

2 Related Work

Hessian similarity in distributed optimization. The similarity assumption has been developed in distributed optimization starting from Shamir et al. [2014], who proposed the communication-efficient Newton-type method DANE. The authors exploited the fact that local objectives are mutually aligned across workers. However, their analysis was limited to quadratic problems. This direction was further developed by accelerated versions of Newton-type methods, including AIDE Reddi et al. [2016], DISCO Zhang and Lin [2015], and GIANT Wang et al. [2018], which improved robustness

and acceleration of approximate Newton-type updates. More recently, similarity became an explicit structural assumption, enabling reduced communication Tian et al. [2022]. The first line of work in this direction focused on the number of *communication rounds*: SONATA Sun et al. [2022] and its accelerated version Tian et al. [2022] were analyzed in the full-participation setting. Kovalev et al. [2022] proposed Accelerated Extragradient, whose complexity bound $\mathcal{O}(\sqrt{\delta/\mu} \log(1/\varepsilon))$ on the number of rounds matches the lower bound of Arjevani and Shamir [2015]. However, since each round in the full-participation setting requires all $n - 1$ clients to transmit their gradients to the server, the number of *exchanged vectors* grows as $\mathcal{O}(n\sqrt{\delta/\mu} \log(1/\varepsilon))$. More recent works measure complexity in terms of the total number of exchanged vectors and reduce it through client sampling, where only a small subset of clients communicates per round. For example, Khaled and Jin [2022] proposed SVRP and Catalyzed SVRP, combining stochastic proximal point evaluations with variance reduction. SVRP attains $\mathcal{O}((n + \delta^2/\mu^2) \log(1/\varepsilon))$ communication complexity, and Catalyzed SVRP improves this to $\mathcal{O}((n + n^{3/4}\sqrt{\delta/\mu}) \log(1/\varepsilon) \log(L/\mu))$ via the Catalyst acceleration framework Lin et al. [2015]. However, both algorithms require μ -strong convexity of each f_i – a stronger assumption than necessary for acceleration. In addition, Catalyzed SVRP pays an extra $\log(L/\mu)$ factor in complexity due to the Catalyst framework.

Later, the authors of Lin et al. [2023] introduced SVRS. Taking into account the complexity gap in Catalyzed SVRP, they applied the KatyushaX framework to obtain an accelerated version named AccSVRS. The algorithm replaces $\nabla f(x_t)$ by the client-sampled estimate $\nabla f_{i_t}(x_t) - g_t$, where $g_t = \nabla f_{i_t}(w_0) - \nabla f(w_0)$ is a variance-reduction correction term computed at a reference point w_0 . AccSVRS achieves an $\mathcal{O}((n + n^{3/4}\sqrt{\delta/\mu}) \log(1/\varepsilon))$ communication bound, matching the corresponding lower bound, requiring strong convexity only of the average f , and removing the $\log(L/\mu)$ factor present in Catalyzed SVRP. Nevertheless, the convergence analysis still relies on μ strong convexity, and periodic calculation of full gradient ∇f i.e., full client participation at the start of each epoch.

No full gradient methods. A classic way to avoid computing the full gradient in finite-sum optimization is to aggregate past gradient information across components. The earliest methods in this line are incremental aggregated gradient methods: IAG, DIAG, PIAG Gurbuzbalaban et al. [2017], Mokhtari et al. [2018], Vanli et al. [2018], which cycle deterministically through components. Building on this, SAG Roux et al. [2012] and SAGA Defazio et al. [2014] introduced randomized component selection, with SAGA correcting the gradient bias present in SAG. However, all these algorithms maintain a table of n past gradients and therefore require $\mathcal{O}(nd)$ memory, which is prohibitive in large-scale or distributed settings; in addition, the deterministic-cycling variants (IAG, DIAG, PIAG) produce biased estimators and were originally analyzed only under strong convexity.

Another line of work uses recursive estimators to achieve $\mathcal{O}(d)$ memory. The classical methods, including SARAH Nguyen et al. [2017] and SPIDER Fang et al. [2018], and their variants - periodically recompute the full gradient ∇f at the start of each epoch as an anchor that resets the variance accumulated over the previous epoch.

More recently, a growing body of work has emerged that uses $\mathcal{O}(d)$ memory while also seeking to avoid full-gradient computations. STORM Cutkosky and Orabona [2020] avoids the anchor via a momentum-based recursive estimator that never requires a full gradient or mega-batch, at the cost of requiring knowledge of the stochastic variance. PAGE Li et al. [2021a] replaces the fixed outer loop by probabilistic switching between a small adjustment and a rare large-batch refresh, achieving the optimal $\mathcal{O}(n + n^{1/2}/\varepsilon^2)$ complexity but still relying on occasional large-batch computations. ZeroSARAH Li et al. [2021b] was proposed as the first variance-reduced method for non-convex finite-sum optimization with zero full-gradient computation, replacing the anchor with an auxiliary momentum sequence. More recently, Medyakov et al. [2025] generalized this viewpoint and argued that full-gradient refreshes are not intrinsic to variance reduction, proposing no-full-gradient variants of both SVRG and SARAH. They keep the classical update rule of these methods intact and replace the full-gradient anchor with different approximation. All of the above works, however, address *centralized* finite-sum optimization and do not exploit second-order similarity between local objectives.

No full gradient methods under similarity. Several recent works explore the interplay between variance reduction without full-gradient computation and hessian similarity structure. In the strongly convex regime, Beznosikov and Takáč [2024] showed that SARAH can be modified to remove full-gradient refreshes via random reshuffling and within-epoch gradient aggregation. Since each reshuffling

pass visits every component exactly once, the accumulated stochastic gradients implicitly serve as a full-gradient estimate. Under δ -Hessian similarity and μ -strong convexity of the average objective, their method achieves communication complexity $\mathcal{O}((nL/\mu + n^2\delta/\mu) \log(1/\varepsilon))$. This analysis is restricted to the centralized setting and to strongly convex objectives, requires each f_i to be convex and L -smooth.

Moving to the non-convex setting, SILVER Oko et al. [2024] builds a single-loop estimator in the SARAH type and extends it to federated learning through local updates, attaining $\mathcal{O}(n + \delta n^{1/2}/\varepsilon^2)$ communicated gradients under Hessian heterogeneity δ , with a matching lower bound. The server must store a per-client state vector resulting in $\mathcal{O}(nd)$ memory, and the federated algorithm is obtained by composing a centralized method with local updates rather than being designed as a distributed algorithm from the start. SPAM Karagulyan et al. [2024] takes a different route: it pairs STORM-type momentum variance reduction on the server with stochastic proximal point evaluations on the clients, also under second-order similarity. It avoids full-gradient computation entirely and attains communication complexity $\mathcal{O}((\delta + \sigma^2)/\varepsilon^2 + \delta\sigma/\varepsilon^3)$, where σ^2 bounds the stochastic gradient variance. However, this rate corresponds to the stochastic regime and does not exploit finite-sum structure as effectively as recursive methods.

3 Setup

3.1 Notation and assumptions

Before presenting theoretical results, we present a list of assumptions.

Assumption 1 (Lower-boundedness). *The function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ attains its minimum (may be not unique), i.e. there exists such x^* that*

$$f(x^*) = \inf_{x \in \mathbb{R}^d} f(x) > -\infty$$

Neural networks tend to have a complex loss landscape [Cybenko, 1989, Nguyen and Hein, 2018]. Since we are motivated by real-world scenarios, we provide convergence results in the non-convex case.

Assumption 2 (bounded hessian dissimilarity). *Local objectives $\{f_i\}_{i=1}^n$ satisfy the δ -second-order similarity definition 1.*

Assumption 2 can be considered as a generalization of smoothness. Indeed, if all f_i are L -smooth, then by triangle inequality $\|\nabla^2 f_i(x) - \nabla^2 f(x)\| \leq 2L$, so L smoothness implies $2L$ similarity of $\{f_i\}_{i=1}^n$. The opposite does not hold in general. As discussed above, if local datasets are sampled i.i.d from a common distribution, $\delta = \mathcal{O}(L/\sqrt{m})$, where m is the local dataset size Hendrikx et al. [2020]. It follows that $\delta \ll L$, because in modern machine learning, datasets contain large number of samples Kaplan et al. [2020].

Notation. Throughout the algorithm and its analysis we fix an integer batch size $b \in \{1, 2, \dots, n\}$. At each epoch, the server partitions the index set $[n] = \{1, \dots, n\}$ into $\lceil n/b \rceil$ disjoint batches of size b .

4 Algorithm

Our setting has only one structural assumption: δ second order similarity 2, and we do not assume the L smoothness of f . This means that classical descent inequalities are not applicable in this case. To circumvent this, we adopt the proximal decomposition of f , introduced by similarity-based methods Accelerated Extragradient, SVRP, SVRS/AccSVRS, as follows:

$$f(w) = (f - f_1)(w) + f_1(w)$$

This suggests a proximal step update

$$w_{t+1} = \text{prox}_{\theta f_1}(w_t - \theta \nabla(f - f_1)(w_t)) = \arg \min_{w \in \mathbb{R}^d} \left\{ \langle \nabla(f - f_1)(w_t), w \rangle + \frac{1}{2\theta} \|w - w_t\|^2 + f_1(w) \right\}.$$

The update is identical to the proximal step in Accelerated Extragradient. However, computing $\nabla(f - f_1)$ requires gradients from all $n - 1$ clients per round. We aim to replace $\nabla(f - f_1)$ by an estimator v .

185 **From SARAH to a no full gradient estimator** A possible approach is the SARAH recursion Nguyen
186 et al. [2017], which builds a low-variance recursive estimator of the form:

$$v_t = v_{t-1} + \left(\nabla(f_i - f_1)(w_t) - \nabla(f_i - f_1)(w_{t-1}) \right). \quad (3)$$

187 But at the beginning of the epoch, the initialization $v_0 = \nabla(f - f_1)(w_0)$ requires the full gradient.
188 Without it, $\|v_t - \nabla(f - f_1)(w_t)\|^2$ does not start from zero. The question becomes how to initialize
189 v_0 in the beginning of each epoch to control variance without full client participation.

190 We construct $v_0^{(s)}$, v_0 at epoch s from gradients that have already evaluated during the previous epoch
191 $s - 1$, following the approach of Beznosikov and Takáč [2024] and Medyakov et al. [2025] in the
192 centralized setup. We adapt it to distributed setting, and apply it to $\nabla(f - f_1)$. In parallel with $v_t^{(s)}$,
193 the server maintains $\tilde{v}_t^{(s)}$ and update it recursively within-epoch running mean of the client's gradients

$$\tilde{v}_{t+1}^{(s)} = \frac{t-1}{t} \tilde{v}_t^{(s)} + \frac{1}{tb} \sum_{i \in B_t^{(s)}} \nabla(f_i - f_1)(w_t^{(s)}). \quad (4)$$

194 Unrolling the recursion gives the closed form.

$$\tilde{v}_{n/b+1}^{(s)} = \frac{1}{n} \sum_{t=1}^{n/b} \sum_{i \in B_t^{(s)}} \nabla(f_i - f_1)(w_t^{(s)}). \quad (5)$$

195 Because batches $B_1^{(s)}, B_2^{(s)}, \dots, B_{\lfloor n/b \rfloor}^{(s)}$ are disjoint, and together cover all n clients, $\tilde{v}_{\lfloor n/b \rfloor+1}^{(s)}$
196 averages exactly one gradient from each client. We then set $v_0^{(s+1)} := \tilde{v}_{\lfloor n/b \rfloor+1}^{(s)}$. The next two
197 lemmas formalize how $\tilde{v}$ controls the variance.

198 **Lemma 1.** *The recursive estimator $v_t^{(s)}$ in 3 satisfies*

$$\|v_t^{(s)} - \nabla(f - f_1)(w_t^{(s)})\|^2 \leq e \|v_0^{(s)} - \nabla(f - f_1)(w_0^{(s)})\|^2 + \frac{18en\delta^2}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2.$$

199 **Lemma 2.** *The initial approximator $v_0^{(s)}$ variance bound is*

$$e \|v_0^{(s)} - \nabla(f - f_1)(w_0^{(s)})\|^2 \leq \frac{16en\delta^2}{b} \sum_{j=0}^{n/b} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2.$$

200 the initial variance of every epoch is bounded by the iterate movement of the previous epoch, so that
201 progress in one epoch pays for the variance of the next.

202 We now describe NFG-SS (NoFullGrad SARAH Similarity), formally described in Algorithm 1. It
203 proceeds in epochs $s = 0, 1, \dots, S$. In the Line 4, at start of the epoch s , server samples disjoint
204 mini-batches of clients of size b : $B_1^{(s)}, B_2^{(s)}, \dots, B_{\lfloor n/b \rfloor}^{(s)}$, that together cover all n workers. At
205 iteration t , server sends model weights $w_t^{(s)}, w_{t-1}^{(s)}$ to workers in $B_t^{(s)}$. These workers compute
206 gradients $\nabla f_i(w_t^{(s)}), \nabla f_i(w_{t-1}^{(s)})$ and send them back to the server, which also computes $\nabla f_1(w_t^{(s)}),$
207 $\nabla f_1(w_{t-1}^{(s)})$. After that, the server updates two recursive estimators $\tilde{v}_{t+1}^{(s)}$ and $v_t^{(s)}$ in Line 8 and 9
208 accordingly, and performs a proximal step on its own local objective f_1 in line 10:

$$w_{t+1}^{(s)} = \text{prox}_{\theta f_1}(w_t^{(s)} - \theta v_t^{(s)}) = \arg \min_{w \in \mathbb{R}^d} \left\{ \langle v_t^{(s)}, w \rangle + \frac{1}{2\theta} \|w - w_t^{(s)}\|^2 + f_1(w) \right\}. \quad (6)$$

209 Crucially, this construction never requires computing the full gradient ∇f .

210 Now we can present the final result of this section: convergence rate and communication complexity
211 of NFG-SS method. The proof is deferred to Appendix B.7.

Algorithm 1 NFG-SS

```

1: Input: Initial point  $w_0^{(0)} \in \mathbb{R}^d$ ; initial gradients  $\tilde{v}_1^{(0)} = 0^d, v_0^{(0)} = 0^d$ 
2: Parameter: Stepsize  $\theta > 0$ , batch size  $b$ 
3: for epochs  $s = 0, 1, \dots, S$  do
4:   Sample batches  $B_1^{(s)}, \dots, B_{\lceil n/b \rceil}^{(s)}$ 
5:    $v_0^{(s)} = v^{(s)}$ 
6:    $w_1^{(s)} = \text{prox}_{\theta f_1}(w_0^{(s)} - \theta v_0^{(s)})$ 
7:   for  $t = 1, 2, \dots, \lceil n/b \rceil$  do
8:     
$$\tilde{v}_{t+1}^{(s)} = \frac{t-1}{t} \tilde{v}_t^{(s)} + \frac{1}{tb} \sum_{i \in B_t^{(s)}} \nabla(f_i - f_1)(w_t^{(s)})$$

9:     
$$v_t^{(s)} = v_{t-1}^{(s)} + \frac{1}{b} \sum_{i \in B_t^{(s)}} (\nabla(f_i - f_1)(w_t^{(s)}) - \nabla(f_i - f_1)(w_{t-1}^{(s)}))$$

10:    
$$w_{t+1}^{(s)} = \arg \min_w \left\{ \langle v_t^{(s)}, w \rangle + \frac{1}{2\theta} \|w - w_t^{(s)}\|^2 + f_1(w) \right\}$$

11:   end for
12:    $w_0^{(s+1)} = w_{\lceil n/b \rceil + 1}^{(s)}$ 
13:    $\tilde{v}_1^{(s+1)} = 0^d$ 
14:    $v^{(s+1)} = \tilde{v}_{\lceil n/b \rceil + 1}^{(s)}$ 
15: end for

```

212 **Theorem 1.** Suppose Assumptions 1 and 2 hold, and define the gradient mapping
213 $G_\theta(w) = \frac{1}{\theta} (w - \text{prox}_{\theta f_1}(w - \theta \nabla(f - f_1)(w)))$. Let Algorithm 1 be run with step size $\theta = \frac{cb}{\delta n}$,
214 where $c^* = \sqrt{\frac{-13 + \sqrt{185}}{272e}}$. Then, to achieve

$$\frac{1}{S(n/b + 1)} \sum_{s=1}^S \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2 \leq \varepsilon^2,$$

215 Algorithm 1 requires at most

$$S = \mathcal{O}\left(\frac{b \delta \Delta_1}{n \varepsilon^2}\right) + \mathcal{O}\left(\frac{n \delta^2 \Delta_2}{b \varepsilon^2}\right)$$

216 epochs and

$$N_{\text{oracle}} = S \cdot n = \mathcal{O}\left(\frac{n \delta \Delta_1}{\varepsilon^2}\right) + \mathcal{O}\left(\frac{n^2 \delta^2 \Delta_2}{b \varepsilon^2}\right)$$

217 oracle calls, where $\Delta_1 = f(w^{(0)}) - f^*$ and $\Delta_2 = \sum_{j=0}^{n/b-1} \|w_{j+1}^{(0)} - w_j^{(0)}\|^2$. In particular, for
218 $b = n^\beta$ with $\beta \in (0, 1]$:

$$N_{\text{oracle}} = \mathcal{O}\left(\frac{n \delta \Delta_1}{\varepsilon^2}\right) + \mathcal{O}\left(\frac{n^{2-\beta} \delta^2 \Delta_2}{\varepsilon^2}\right).$$

219 The first term $\mathcal{O}(n \delta \Delta_1 / \varepsilon^2)$ is independent of the b and comes from the variance accumula-
220 tion over epoch because of the SARAH estimator. It matches the well-known oracle complexity
221 $\mathcal{O}(n L \Delta / \varepsilon^2)$, established for no full gradient method No Full Grad SARAH Medyakov et al. [2025]
222 with the smoothness constant replaced by second order similarity constant δ . The second term
223 $\mathcal{O}(n^2 \delta^2 \Delta_2 / (b \varepsilon^2))$ emerges because the initial gradient estimation $v^{(s+1)}$ in Line 13 is constructed
224 from gradients computed at the iterates of the previous epoch.

Two prior works, Shuffled SARAH and SILVER, were discussed in No full gradient methods under similarity part of Related Work. Firstly, like Algorithm 1, Shuffled SARAH relies on running mean accumulator $\tilde{v}$ during the epoch to avoid full gradient calculation. But the difference is that they aggregate ∇f_i directly, whereas we apply it to the difference $\nabla(f - f_i)$, and then update parameters with a proximal step. This formulation allows the δ similarity assumption to enter the convergence bound, and to write down the descent step. Second, Shuffled SARAH is restricted to centralized learning, and processes one component in one iteration, while our method aggregates a batch of clients in one time. Thirdly, their analysis is conducted in the strongly convex regime with a convergence rate $\mathcal{O}((nL/\mu + n^2\delta/\mu)\log(1/\varepsilon))$. The n^2 dependence is suboptimal: the optimal rate for similarity-based methods in their setting is $\mathcal{O}(n + n^{3/4}\sqrt{\delta/\mu})$, proved in Lin et al. [2023].

Closer in spirit, SILVER Oko et al. [2024] also targets distributed non-convex optimization with a SARAH-type estimator under similarity, reaching a communication complexity of $\mathcal{O}(n + \delta n^{1/2}/\varepsilon^2)$ in the number of transmitted gradients under Hessian heterogeneity δ . However, SILVER maintains a per-client state vector on the server, costing $\mathcal{O}(nd)$ memory; in contrast, the running-mean accumulator $\tilde{v}_t^{(s)}$ keeps the server-side memory at $\mathcal{O}(d)$, independent of n , which is essential when n is large.

5 Experiments

To complement the theoretical analysis, we conduct a controlled empirical study on image classification. The goal of this section is to evaluate whether the proposed no-full-gradient estimator can preserve practical convergence while avoiding full-gradient refreshes.

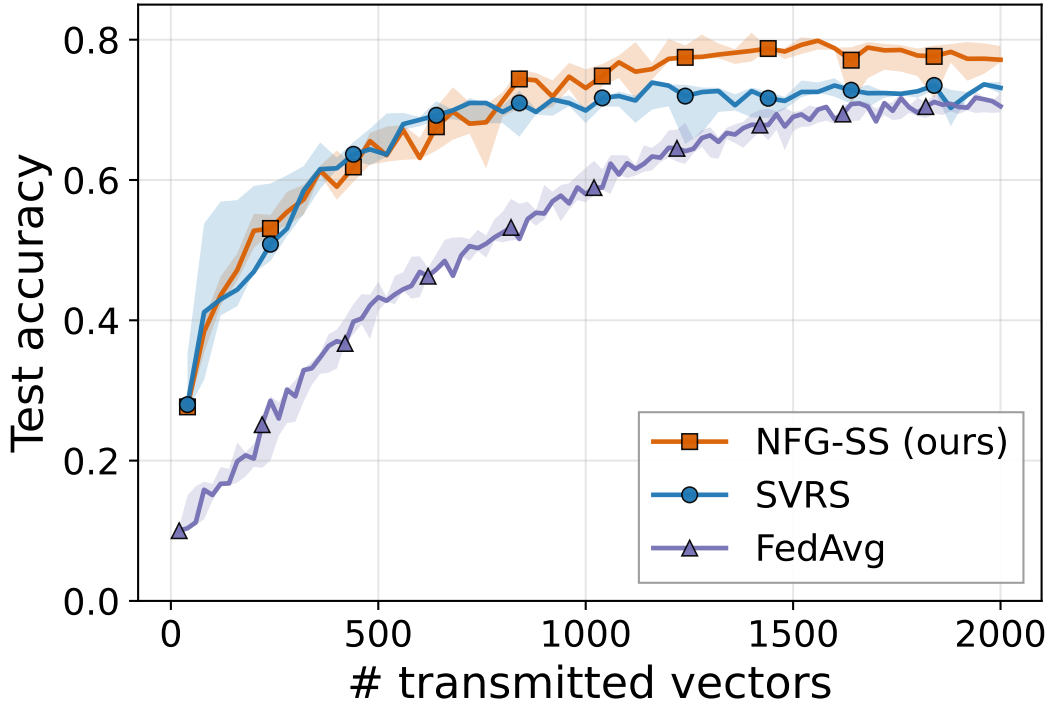


Figure 1: Comparison of NFG-SS, SVRS, and FedAvg on CIFAR-10.

Experimental Setup. We train RESNET-18 [He et al., 2016] on CIFAR-10 [Krizhevsky et al., 2009] using the cross-entropy loss. The data is distributed uniformly at random across $n = 11$ nodes: one node is designated as the server-side objective f_1 , while the remaining ten nodes act as clients $f_2, \dots, f_{11}$. We report training loss and test accuracy as the main performance metrics.

We compare NFG-SS against SVRS [Lin et al., 2023] and FedAvg [McMahan et al., 2017]. All methods are evaluated under the same data partition. For NFG-SS and SVRS, we also use the same inexact proximal solver and the same number of outer epochs. We allocate $1/2$ of the total training

data to the server-side objective f_1 and distribute the remaining data uniformly across the clients. This setup makes the proximal subproblem sufficiently informative and allows us to isolate the effect of replacing full-gradient refreshes with the running-mean estimator.

For SVRS, we follow the original method and compute the exact full-gradient reference $\frac{1}{n} \sum_i \nabla(f_i - f_1)(w_0)$ at the beginning of each epoch by querying every client over its local dataset. In contrast, NFG-SS initializes its estimator using gradients accumulated during the previous epoch and therefore avoids full-gradient computation at a common iterate. FedAvg is run in a communication-matched regime, where the total number of transmitted vectors over training is set equal to that of NFG-SS.

In all experiments, we use client batch size $b = 1$. The proximal step on f_1 in both NFG-SS and SVRS is approximated by the same inexact solver. Details of the solver, hyperparameter tuning, and implementation are provided in Appendix A.

Experiment Results. Figure 1 reports test accuracy as a function of the number of transmitted vectors. NFG-SS achieves higher test accuracy than SVRS under the same communication budget. Moreover, it reaches the SVRS accuracy plateau substantially earlier in training, suggesting that the running-mean estimator $\tilde{v}_t^{(s)}$ provides effective variance control without requiring a full-gradient refresh at the start of each epoch. Communication-matched FedAvg remains below both variance-reduced methods, which highlights the importance of variance reduction under a fixed communication budget. Overall, these results support the main empirical message of the paper: in this controlled setting, full-gradient refreshes can be replaced by the proposed no-full-gradient estimator without degrading the convergence.

6 Conclusion

We proposed NFG-SS, a distributed variance-reduced method for non-convex optimization under second-order similarity. The method combines a SARA-type recursive estimator with a running-mean accumulator built from gradients already computed during the previous epoch. This construction removes the need for full-gradient refreshes while preserving the dependence on the similarity parameter δ .

To the best of our knowledge, NFG-SS is the first distributed method that exploits second-order similarity without any full client communication at a common iterate. Existing similarity-based methods rely on full-gradient computations or synchronized full-participation rounds, whereas no-full-gradient methods are mainly developed for centralized finite-sum optimization and do not benefit from the regime $\delta \ll L$. Our work bridges these two directions by showing that similarity can be used even when full communication is completely avoided.

We prove convergence guarantees for non-convex objectives under second-order similarity and show that the resulting complexity depends on δ rather than on the smoothness constant L . Experiments on CIFAR-10 with RESNET-18 support the theoretical findings and demonstrate that the proposed running-mean estimator can replace full-gradient refreshes without degrading practical convergence.

References

- Yossi Arjevani and Ohad Shamir. Communication complexity of distributed convex learning and optimization. *Advances in neural information processing systems*, 28, 2015.
- Aleksandr Beznosikov and Martin Takáč. Random-reshuffled sarah does not need full gradient computations. *Optimization Letters*, 18(3):727–749, 2024.
- Léon Bottou, Frank E. Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018.
- Ashok Cutkosky and Francesco Orabona. Momentum-based variance reduction in non-convex sgd, 2020. URL <https://arxiv.org/abs/1905.10018>.
- George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of control, signals and systems*, 2(4):303–314, 1989.

- 298 Aaron Defazio, Francis Bach, and Simon Lacoste-Julien. Saga: A fast incremental gradient method
299 with support for non-strongly convex composite objectives. *Advances in neural information*
300 *processing systems*, 27, 2014.
- 301 Ofer Dekel, Ran Gilad-Bachrach, Ohad Shamir, et al. Optimal distributed online prediction using
302 mini-batches. *Journal of Machine Learning Research*, 13(1), 2012.
- 303 Cong Fang, Chris Junchi Li, Zhouchen Lin, and Tong Zhang. Spider: Near-optimal non-convex
304 optimization via stochastic path-integrated differential estimator. *Advances in neural information*
305 *processing systems*, 31, 2018.
- 306 Mert Gurbuzbalaban, Asuman Ozdaglar, and Pablo A Parrilo. On the convergence rate of incremental
307 aggregated gradient algorithms. *SIAM Journal on Optimization*, 27(2):1035–1048, 2017.
- 308 Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image
309 recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*,
310 pages 770–778, 2016.
- 311 Hadrien Hendrikx, Lin Xiao, Sebastien Bubeck, Francis Bach, and Laurent Massoulié. Statisti-
312 cally preconditioned accelerated gradient method for distributed optimization. In *International*
313 *conference on machine learning*, pages 4203–4227. PMLR, 2020.
- 314 Peter Kairouz and H Brendan McMahan. Advances and open problems in federated learning.
315 *Foundations and trends in machine learning*, 14(1-2):1–210, 2021.
- 316 Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child,
317 Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models.
318 *arXiv preprint arXiv:2001.08361*, 2020.
- 319 Avetik Karagulyan, Egor Shulgin, Abdurakhmon Sadiev, and Peter Richtárik. Spam: Stochastic
320 proximal point method with momentum variance reduction for non-convex cross-device federated
321 learning, 2024. URL <https://arxiv.org/abs/2405.20127>.
- 322 Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and
323 Ananda Theertha Suresh. Scaffold: Stochastic controlled averaging for federated learning. In
324 *International conference on machine learning*, pages 5132–5143. PMLR, 2020.
- 325 Ahmed Khaled and Chi Jin. Faster federated optimization under second-order similarity. *arXiv*
326 *preprint arXiv:2209.02257*, 2022.
- 327 Ahmed Khaled, Konstantin Mishchenko, and Peter Richtárik. Tighter theory for local sgd on identical
328 and heterogeneous data. In *International conference on artificial intelligence and statistics*, pages
329 4519–4529. PMLR, 2020.
- 330 Jakub Konečný, H Brendan McMahan, Felix X Yu, Peter Richtárik, Ananda Theertha Suresh, and
331 Dave Bacon. Federated learning: Strategies for improving communication efficiency. *arXiv*
332 *preprint arXiv:1610.05492*, 2016.
- 333 Dmitry Kovalev, Aleksandr Beznosikov, Ekaterina Borodich, Alexander Gasnikov, and Gesualdo
334 Scutari. Optimal gradient sliding and its application to optimal distributed optimization under
335 similarity. *Advances in Neural Information Processing Systems*, 35:33494–33507, 2022.
- 336 Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009.
- 337 Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith.
338 Federated optimization in heterogeneous networks. *Proceedings of Machine learning and systems*,
339 2:429–450, 2020.
- 340 Zhize Li, Hongyan Bao, Xiangliang Zhang, and Peter Richtárik. Page: A simple and optimal
341 probabilistic gradient estimator for nonconvex optimization. In *International conference on*
342 *machine learning*, pages 6286–6295. PMLR, 2021a.
- 343 Zhize Li, Slavomír Hanzely, and Peter Richtárik. Zerosarah: Efficient nonconvex finite-sum opti-
344 mization with zero full gradient computation. *arXiv preprint arXiv:2103.01447*, 2021b.

- 345 Dachao Lin, Yuze Han, Haishan Ye, and Zhihua Zhang. Stochastic distributed optimization under
346 average second-order similarity: Algorithms and analysis. *Advances in Neural Information*
347 *Processing Systems*, 36:1849–1862, 2023.
- 348 Hongzhou Lin, Julien Mairal, and Zaid Harchaoui. A universal catalyst for first-order optimization.
349 *Advances in neural information processing systems*, 28, 2015.
- 350 Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas.
351 Communication-efficient learning of deep networks from decentralized data. In *Artificial intelli-*
352 *gence and statistics*, pages 1273–1282. Pmlr, 2017.
- 353 Daniil Medyakov, Gleb Molodtsov, Savelii Chezhegov, Alexey Rebrov, and Aleksandr Beznosikov.
354 Variance reduction methods do not need to compute full gradients: Improved efficiency through
355 shuffling. *arXiv preprint arXiv:2502.14648*, 2025.
- 356 Aryan Mokhtari, Mert Gurbuzbalaban, and Alejandro Ribeiro. Surpassing gradient descent provably:
357 A cyclic incremental method with linear convergence rate. *SIAM Journal on Optimization*, 28(2):
358 1420–1447, 2018.
- 359 Lam M Nguyen, Jie Liu, Katya Scheinberg, and Martin Takáč. Sarah: A novel method for machine
360 learning problems using stochastic recursive gradient. In *International conference on machine*
361 *learning*, pages 2613–2621. PMLR, 2017.
- 362 Quynh Nguyen and Matthias Hein. Optimization landscape and expressivity of deep cnns. In
363 *International conference on machine learning*, pages 3730–3739. PMLR, 2018.
- 364 Kazusato Oko, Shunta Akiyama, Denny Wu, Tomoya Murata, and Taiji Suzuki. SILVER: Single-loop
365 variance reduction and application to federated learning. In *Forty-first International Conference on*
366 *Machine Learning*, 2024. URL <https://openreview.net/forum?id=p0gMluzEIH>.
- 367 Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito,
368 Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. Automatic differentiation in
369 pytorch. 2017.
- 370 Sashank J Reddi, Jakub Konečný, Peter Richtárik, Barnabás Póczos, and Alex Smola. Aide: Fast and
371 communication efficient distributed optimization. *arXiv preprint arXiv:1608.06879*, 2016.
- 372 Nicolas Roux, Mark Schmidt, and Francis Bach. A stochastic gradient method with an exponential
373 convergence _rate for finite training sets. *Advances in neural information processing systems*, 25,
374 2012.
- 375 Ohad Shamir, Nati Srebro, and Tong Zhang. Communication-efficient distributed optimization using
376 an approximate newton-type method. In *International conference on machine learning*, pages
377 1000–1008. PMLR, 2014.
- 378 Sebastian U Stich. Local sgd converges fast and communicates little. *arXiv preprint arXiv:1805.09767*,
379 2018.
- 380 Ying Sun, Gesualdo Scutari, and Amir Daneshmand. Distributed optimization based on gradient
381 tracking revisited: Enhancing convergence rate via surrogation. *SIAM Journal on Optimization*, 32
382 (2):354–385, 2022.
- 383 Ye Tian, Gesualdo Scutari, Tianyu Cao, and Alexander Gasnikov. Acceleration in distributed
384 optimization under similarity. In *International Conference on Artificial Intelligence and Statistics*,
385 pages 5721–5756. PMLR, 2022.
- 386 N Denizcan Vanli, Mert Gurbuzbalaban, and Asuman Ozdaglar. Global convergence rate of proximal
387 incremental aggregated gradient methods. *SIAM Journal on Optimization*, 28(2):1282–1300, 2018.
- 388 Joost Verbraeken, Matthijs Wolting, Jonathan Katzy, Jeroen Kloppenburg, Tim Verbelen, and Jan S
389 Rellermeyer. A survey on distributed machine learning. *Acm computing surveys (csur)*, 53(2):
390 1–33, 2020.

- 391 Shusen Wang, Fred Roosta, Peng Xu, and Michael W Mahoney. Giant: Globally improved approx-
392 imate newton method for distributed optimization. *Advances in neural information processing*
393 *systems*, 31, 2018.
- 394 Blake E Woodworth, Kumar Kshitij Patel, and Nati Srebro. Minibatch vs local sgd for heterogeneous
395 distributed learning. *Advances in Neural Information Processing Systems*, 33:6281–6292, 2020.
- 396 Yuchen Zhang and Xiao Lin. Disco: Distributed optimization for self-concordant empirical loss.
397 In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on*
398 *Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 362–370,
399 Lille, France, 07–09 Jul 2015. PMLR.
- 400 Martin Zinkevich, Markus Weimer, Lihong Li, and Alex Smola. Parallelized stochastic gradient
401 descent. In *Advances in Neural Information Processing Systems*, volume 23, 2010.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: see Paragraph 1 and Section 4.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: all the assumptions are introduced in Section 3, and further limitations are discussed at the end of Section 4.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: see Sections 3, 4, B.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: see Sections 5, A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: this paper does not have a centralized accessible code space.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: see Sections 5, A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: since the experiments are more of a theoretical verification, we have no statistical effects associated with running the experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: see Section A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: the paper follows the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: that there is no societal impact of the work performed.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: the paper poses no such risks.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: see Sections 5, A.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: the paper does not release new assets.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: the paper does not involve crowdsourcing nor research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: the paper does not involve crowdsourcing nor research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

711 Question: Does the paper describe the usage of LLMs if it is an important, original, or
712 non-standard component of the core methods in this research? Note that if the LLM is used
713 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
714 scientific rigor, or originality of the research, declaration is not required.

715 Answer: [N/A]

716 Justification: the core method development in this paper does not involve LLMs as any
717 important, original, or non-standard components.

718 Guidelines:

- 719 • The answer [N/A] means that the core method development in this research does not
720 involve LLMs as any important, original, or non-standard components.
- 721 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
722 be described.

723 A Experimental details.

724 **Hardware Details.** All experiments were implemented in Python using the PyTorch deep learning
725 framework [Paszke et al., 2017]. The computational infrastructure consisted of a server equipped
726 with an Intel Xeon Gold 6342 CPU and two NVIDIA A100 40 GB GPUs. The total wall-clock time
727 for the full set of experiments was approximately 60 hours.

728 **Inexact Proximal Subproblem.** At every outer step we must evaluate

$$w^* \approx \text{prox}_{\theta f_1}(w_{\text{out}} - \theta v) = \arg \min_w \left\{ \frac{1}{2\theta} \|w - z\|^2 + f_1(w) \right\}, \quad z := w_{\text{out}} - \theta v, \quad (7)$$

729 where w_{out} is the current outer iterate, v is the auxiliary direction maintained by the outer loop, $\theta > 0$
730 is the proximal step-size, and f_1 is accessible only through stochastic gradients on a server-side
731 mini-batch ξ . The iterate is warm-started at $w_0 = z$, which absorbs the linear term $\langle \theta v, \cdot \rangle$ into the
732 initialization, and at iteration t we form the search direction

$$d_t = (w_t - w_{\text{out}}) + \theta \nabla f_1(w_t; \xi_t), \quad (8)$$

733 i.e. the gradient of $\Psi(w) = \frac{1}{2} \|w - w_{\text{out}}\|^2 + \theta f_1(w)$ at w_t . Since the linear term is encoded only by
734 the warm-start, a small number of inner epochs (we determine here 4 passes over the server loader
735 for achieving the proper solution) is critical: the iterate stays in the v -shifted neighbourhood of z , and
736 the bias of running (8) to convergence is negligible, even though its fixed point would otherwise drift
737 towards $\text{prox}_{\theta f_1}(w_{\text{out}})$.

738 **Inner Update.** Direction d_t is smoothed by a heavy-ball term, $d_t \leftarrow \mu p_{t-1} + d_t$ with $p_t = d_t$ and
739 $\mu = 0.9$. The parameters are then updated either by SGD with decoupled weight decay $\gamma_{\text{wd}} = 0.1$,
740 $w_{t+1} = (1 - \gamma_{\text{wd}} \eta_t) w_t - \eta_t d_t$. The step-size starts at $\eta_0 = \kappa / (2(1 + \theta L_1))$, where L_1 bounds the
741 Lipschitz constant of ∇f_1 and κ is a tunable multiplier. The inner loop terminates either after a
742 prescribed budget of passes or as soon as $\|d_t\| / \|d_0\| < \varepsilon_{\text{stop}}$.

743 **Hyperparameters.** For every method we tune the hyperparameters listed in Table 1 via a hand-crafted
744 grid, using grid-search. The score of a trial is its best validation accuracy. We then evaluate the best
745 configuration on the held-out test set.

Method	Hyperparameter	Search values
NFG-SS, SVRS	step size θ	[0.1, 0.15, 0.2, 0.25, 0.5, 1.0]
	inner-prox epochs K	[4, 5, 6]
	inner-prox momentum μ	[0.0, 0.5, 0.9, 0.95]
	inner-prox weight decay γ_{wd}	[0.0, 0.01, 0.05, 0.1]
	inner-prox learning rate factor κ	[0.1, 0.15, 0.2, 0.25, 0.5, 1.0]
FedAvg	local learning rate	[0.005, 0.007, 0.01, 0.03, 0.05]

Table 1: Hyperparameter search grids.

746 **Design features.** As a practical modification without theoretical guarantees, we introduce a schedule
747 for the cumulative gradient difference v to make the proximal subproblem more stable when the inner
748 loop is short and v is noisy. As the inner loop progresses, the weight on v is gradually reduced, which
749 dampens overshooting and limits the effect of an imperfect v , keeping the update closer to the current
750 outer iterate w_{outer} . This schedule therefore balances aggressiveness and stability: the early steps
751 exploit v , while the later steps stabilize the trajectory and improve convergence.

752 B Proofs

753 B.1 General Inequalities

754 Firstly, we will list several standard inequalities used throughout the proofs. For all vectors
755 $x, y, \{x_i\} \in \mathbb{R}^d$, for all scalars $\alpha, \beta > 0$:

$$2\langle x, y \rangle \leq \frac{\|x\|^2}{\alpha} + \alpha\|y\|^2, \quad (\text{Scalar})$$

$$2\langle x, y \rangle = \|x + y\|^2 - \|x\|^2 - \|y\|^2, \quad (\text{Norm})$$

$$\|x + y\|^2 \leq (1 + \beta)\|x\|^2 + (1 + \frac{1}{\beta})\|y\|^2, \quad (\text{Quad})$$

$$\left\| \sum_{i=1}^n x_i \right\|^2 \leq n \sum_{i=1}^n \|x_i\|^2 \quad (\text{Cauchy-Schwarz}), \quad (\text{CS})$$

(9)

756 B.2 Proof of Lemma 3

Lemma 3.

$$f(w_{t+1}^{(s)}) \leq f(w_t^{(s)}) + \left(-\frac{1}{4\theta} + \delta\right) \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + \theta \|v_t^{(s)} - \nabla(f - f_1)(w_t^{(s)})\|^2$$

757 *Proof.* Start from

$$f(w_{t+1}^{(s)}) - f(w_t^{(s)}) = \int_0^1 \langle \nabla f(w_t^{(s)} + \tau(w_{t+1}^{(s)} - w_t^{(s)})), w_{t+1}^{(s)} - w_t^{(s)} \rangle d\tau$$

758 And the same for f_1

$$f_1(w_{t+1}^{(s)}) - f_1(w_t^{(s)}) = \int_0^1 \langle \nabla f_1(w_t^{(s)} + \tau(w_{t+1}^{(s)} - w_t^{(s)})), w_{t+1}^{(s)} - w_t^{(s)} \rangle d\tau$$

759 Considering these two inequalities together, we obtain

$$f(w_{t+1}^{(s)}) - f(w_t^{(s)}) = \int_0^1 \langle \nabla(f - f_1)(w_t^{(s)} + \tau(w_{t+1}^{(s)} - w_t^{(s)})), w_{t+1}^{(s)} - w_t^{(s)} \rangle d\tau + f_1(w_{t+1}^{(s)}) - f_1(w_t^{(s)}).$$

760 By the proximal step definition:

$$f_1(w_{t+1}^{(s)}) + \langle v_t^{(s)}, w_{t+1}^{(s)} - w_t^{(s)} \rangle + \frac{1}{2\theta} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 \leq f_1(w_t^{(s)}),$$

761 Hence

$$f_1(w_{t+1}^{(s)}) - f_1(w_t^{(s)}) \leq -\frac{1}{2\theta} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 - \langle v_t^{(s)}, w_{t+1}^{(s)} - w_t^{(s)} \rangle.$$

762 Then, adding and subtracting $\nabla(f - f_1)(w_t^{(s)})$ as smart zero, we get

$$\begin{aligned} f(w_{t+1}^{(s)}) - f(w_t^{(s)}) &\leq -\frac{1}{2\theta} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 \\ &\quad + \int_0^1 \langle \nabla(f - f_1)(w_t^{(s)} + \tau(w_{t+1}^{(s)} - w_t^{(s)})) - \nabla(f - f_1)(w_t^{(s)}), w_{t+1}^{(s)} - w_t^{(s)} \rangle d\tau \\ &\quad + \langle \nabla(f - f_1)(w_t^{(s)}) - v_t^{(s)}, w_{t+1}^{(s)} - w_t^{(s)} \rangle. \end{aligned}$$

763 For the integral term, by δ similarity 1:

$$\begin{aligned} &\int_0^1 \langle \nabla(f - f_1)(w_t^{(s)} + \tau(w_{t+1}^{(s)} - w_t^{(s)})) - \nabla(f - f_1)(w_t^{(s)}), w_{t+1}^{(s)} - w_t^{(s)} \rangle d\tau \\ &\leq \int_0^1 \delta \|\tau(w_{t+1}^{(s)} - w_t^{(s)})\| \cdot \|w_{t+1}^{(s)} - w_t^{(s)}\| d\tau \leq \delta \|w_{t+1}^{(s)} - w_t^{(s)}\|^2. \end{aligned}$$

764 For the inner product term, by Scalar

$$\begin{aligned} f(w_{t+1}^{(s)}) - f(w_t^{(s)}) &\leq -\frac{1}{2\theta} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + \frac{\alpha}{2} \|\nabla(f-f_1)(w_t^{(s)}) - v_t^{(s)}\|^2 \\ &\quad + \frac{1}{2\alpha} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + \delta \|w_{t+1}^{(s)} - w_t^{(s)}\|^2. \end{aligned}$$

765 Setting $\alpha = 2\theta$:

$$f(w_{t+1}^{(s)}) - f(w_t^{(s)}) \leq \left(-\frac{1}{4\theta} + \delta\right) \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + \theta \|\nabla(f-f_1)(w_t^{(s)}) - v_t^{(s)}\|^2. \quad \square$$

766 B.3 Proof of Lemma 1

767 **Lemma 4** (Lemma 1). For $t = 0, \dots, n/b$:

$$\|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2 \leq e \|v_0^{(s)} - \nabla(f-f_1)(w_0^{(s)})\|^2 + \frac{18e\delta^2 n}{b} \sum_{j=0}^{t-1} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2.$$

768 *Proof.* By the SARAH recursion and adding and subtracting $\nabla(f-f_1)(w_{t-1}^{(s)})$:

$$\begin{aligned} &\|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2 \\ &= \left\| \left[v_{t-1}^{(s)} - \nabla(f-f_1)(w_{t-1}^{(s)}) \right] + \left[\frac{1}{b} \sum_{i \in B_t^{(s)}} \left(\nabla(f_i-f_1)(w_t^{(s)}) - \nabla(f_i-f_1)(w_{t-1}^{(s)}) \right) \right. \right. \\ &\quad \left. \left. - \left(\nabla(f-f_1)(w_t^{(s)}) - \nabla(f-f_1)(w_{t-1}^{(s)}) \right) \right] \right\|^2 \\ &\stackrel{(\text{Quad})}{\leq} \left(1 + \frac{1}{\alpha} \right) \|v_{t-1}^{(s)} - \nabla(f-f_1)(w_{t-1}^{(s)})\|^2 + (\alpha + 1) A_{B_t^{(s)}}(w_t^{(s)}, w_{t-1}^{(s)}), \end{aligned}$$

769 where

$$\begin{aligned} A_{B_t^{(s)}}(w_t^{(s)}, w_{t-1}^{(s)}) &:= \left\| \frac{1}{b} \sum_{i \in B_t^{(s)}} \left(\nabla(f_i-f_1)(w_t^{(s)}) - \nabla(f_i-f_1)(w_{t-1}^{(s)}) \right) \right. \\ &\quad \left. - \left(\nabla(f-f_1)(w_t^{(s)}) - \nabla(f-f_1)(w_{t-1}^{(s)}) \right) \right\|^2. \end{aligned}$$

770 We now bound $A_{B_t^{(s)}}$ deterministically. Denote

$$Y_i := \nabla(f_i-f_1)(w_t^{(s)}) - \nabla(f_i-f_1)(w_{t-1}^{(s)}), \quad \bar{Y} := \frac{1}{n} \sum_{i=1}^n Y_i = \nabla(f-f_1)(w_t^{(s)}) - \nabla(f-f_1)(w_{t-1}^{(s)}).$$

771 Then

$$A_{B_t^{(s)}} = \left\| \frac{1}{b} \sum_{i \in B_t^{(s)}} (Y_i - \bar{Y}) \right\|^2 \stackrel{(\text{CS})}{\leq} \frac{1}{b} \sum_{i \in B_t^{(s)}} \|Y_i - \bar{Y}\|^2.$$

772 For each summand, by (Quad) with $\beta = 1/2$ and using that $\nabla(f_i-f_1)$ is 2δ -Lipschitz and $\nabla(f-f_1)$ is δ -Lipschitz by Assumption 1:

$$\|Y_i - \bar{Y}\|^2 \leq \frac{3}{2} \|Y_i\|^2 + 3 \|\bar{Y}\|^2 \leq 6\delta^2 \|w_t^{(s)} - w_{t-1}^{(s)}\|^2 + 3\delta^2 \|w_t^{(s)} - w_{t-1}^{(s)}\|^2 = 9\delta^2 \|w_t^{(s)} - w_{t-1}^{(s)}\|^2.$$

774 Since $|B_t^{(s)}| = b$, we obtain $A_{B_t^{(s)}} \leq 9\delta^2 \|w_t^{(s)} - w_{t-1}^{(s)}\|^2$. Setting $\alpha = n/b$ and unrolling from t
775 down to 0:

$$\begin{aligned} \|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2 &\leq \left(1 + \frac{b}{n} \right)^{n/b} \|v_0^{(s)} - \nabla(f-f_1)(w_0^{(s)})\|^2 + \left(\frac{n}{b} + 1 \right) \left(1 + \frac{b}{n} \right)^{n/b} \sum_{j=1}^t 9\delta^2 \|w_j^{(s)} - w_{j-1}^{(s)}\|^2 \\ &\leq e \|v_0^{(s)} - \nabla(f-f_1)(w_0^{(s)})\|^2 + \frac{18e n \delta^2}{b} \sum_{j=0}^{t-1} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2. \quad \square \end{aligned}$$

776 **B.4 Proof of Lemma 2**

Lemma 5 (Lemma 2).

$$e \|v_0^{(s)} - \nabla(f - f_1)(w_0^{(s)})\|^2 \leq \frac{16e n \delta^2}{b} \sum_{j=0}^{n/b} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2$$

777 *Proof.* By definition of $v^{(s)} = \tilde{v}_{\lceil n/b \rceil + 1}^{(s)}$, and for $\tilde{v}_{\lceil n/b \rceil + 1}^{(s)}$ applying 5

$$\begin{aligned} e \|v_0^{(s)} - \nabla(f - f_1)(w_0^{(s)})\|^2 &= e \|v^{(s)} - \nabla(f - f_1)(w_0^{(s)})\|^2 \\ &= e \left\| \frac{1}{b} \frac{1}{n/b} \sum_{t=1}^{n/b} \frac{1}{b} \sum_{i \in B_t} \nabla(f_i - f_1)(w_t^{(s-1)}) - \frac{1}{n} \sum_{i=1}^n \nabla(f_i - f_1)(w_0^{(s)}) \right\|^2. \end{aligned}$$

778 Replace $\sum_{i=1}^n$ with double sum $\sum_{t=1}^{n/b} \sum_{B^t}$

$$\leq \frac{e}{n^2} \left\| \sum_{t=1}^{n/b} \sum_{i \in B_t} \left(\nabla(f_i - f_1)(w_t^{(s-1)}) - \nabla(f_i - f_1)(w_0^{(s)}) \right) \right\|^2.$$

779 applying CS to the outer sum over batches (n/b terms).

$$\leq \frac{e}{n^2} \cdot \frac{n}{b} \sum_{t=1}^{n/b} \left\| \sum_{i \in B_t} \left(\nabla(f_i - f_1)(w_t^{(s-1)}) - \nabla(f_i - f_1)(w_0^{(s)}) \right) \right\|^2.$$

780 Applying CS inequality to the inner sum over B_t (b terms):

$$\leq \frac{e}{n} \sum_{t=1}^{n/b} \sum_{i \in B_t} \|\nabla(f_i - f_1)(w_t^{(s-1)}) - \nabla(f_i - f_1)(w_0^{(s)})\|^2.$$

781 By the 2δ -Lipschitzness of each $\nabla(f_i - f_1)$:

$$\leq \frac{4e \delta^2}{n} \sum_{t=1}^{n/b} \sum_{i \in B_t} \|w_t^{(s-1)} - w_0^{(s)}\|^2 = \frac{4e \delta^2 b}{n} \sum_{t=1}^{n/b} \|w_t^{(s-1)} - w_0^{(s)}\|^2.$$

782 We add smart zero via $w_0^{(s-1)}$ and use Quad with $\beta = 1$.

$$\leq \frac{8e \delta^2 b}{n} \sum_{t=1}^{n/b} \|w_0^{(s-1)} - w_0^{(s)}\|^2 + \frac{8e \delta^2 b}{n} \sum_{t=1}^{n/b} \|w_t^{(s-1)} - w_0^{(s-1)}\|^2.$$

783 Since the first term does not depend on t , the sum contributes a factor of n/b :

$$\leq 8e \delta^2 \|w_0^{(s-1)} - w_0^{(s)}\|^2 + \frac{8e \delta^2 b}{n} \sum_{t=1}^{n/b} \|w_t^{(s-1)} - w_0^{(s-1)}\|^2.$$

784 Now we telescope each term. Since $w_0^{(s)} = w_{n/b}^{(s-1)}$:

$$\|w_0^{(s-1)} - w_0^{(s)}\|^2 = \left\| \sum_{j=0}^{n/b-1} (w_{j+1}^{(s-1)} - w_j^{(s-1)}) \right\|^2 \stackrel{CS}{\leq} \frac{n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2,$$

785 and for each t :

$$\|w_t^{(s-1)} - w_0^{(s-1)}\|^2 = \left\| \sum_{j=0}^{t-1} (w_{j+1}^{(s-1)} - w_j^{(s-1)}) \right\|^2 \stackrel{CS}{\leq} t \sum_{j=0}^{t-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2 \leq \frac{n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2.$$

786 Substituting: the first term gives

$$8e\delta^2 \cdot \frac{n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2 = \frac{8e\delta^2 n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2,$$

787 and the second:

$$\frac{8e\delta^2 b}{n} \cdot \frac{n}{b} \cdot \frac{n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2 = \frac{8e\delta^2 n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2.$$

788 Combining:

$$e\|v_0^{(s)} - \nabla(f-f_1)(w_0^{(s)})\|^2 \leq \frac{16e\delta^2 n}{b} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2. \quad \square$$

789 B.5 Proof of lemma 6

790 **Lemma 6.** By definition let's take $G_\theta(w) = \frac{1}{\theta}(w - \text{prox}_{\theta f_1}(w - \theta \nabla(f-f_1)(w)))$

$$\theta^2 \|G_\theta(w_t^{(s)})\|^2 \leq 2\|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + 2\theta^2 \|\nabla(f-f_1)(w_t^{(s)}) - v_t^{(s)}\|^2$$

791 *Proof.* By definition of G_θ and adding smart zero as $w_{t+1}^{(s)}$:

$$\begin{aligned} \theta \|G_\theta(w_t^{(s)})\| &= \|w_t^{(s)} - \text{prox}_{\theta f_1}(w_t^{(s)} - \theta \nabla(f-f_1)(w_t^{(s)}))\| \\ &\leq \|w_t^{(s)} - w_{t+1}^{(s)} + w_{t+1}^{(s)} - \text{prox}_{\theta f_1}(w_t^{(s)} - \theta \nabla(f-f_1)(w_t^{(s)}))\| \\ &\leq \|w_t^{(s)} - w_{t+1}^{(s)}\| + \|\text{prox}_{\theta f_1}(w_t^{(s)} - \theta v_t^{(s)}) - \text{prox}_{\theta f_1}(w_t^{(s)} - \theta \nabla(f-f_1)(w_t^{(s)}))\|, \end{aligned}$$

792 where we substituted the update step of Line 10 of Algorithm 1 is equivalently

793 $\text{prox}_{\theta f_1}(w_t^{(s)} - \theta v_t^{(s)})$. By nonexpansiveness of prox:

$$\theta \|G_\theta(w_t^{(s)})\| \leq \|w_{t+1}^{(s)} - w_t^{(s)}\| + \theta \|\nabla(f-f_1)(w_t^{(s)}) - v_t^{(s)}\|$$

794

$$\theta^2 \|G_\theta(w_t^{(s)})\|^2 \stackrel{\text{(Scalar)}}{\leq} 2\|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + 2\theta^2 \|\nabla(f-f_1)(w_t^{(s)}) - v_t^{(s)}\|^2 \quad \square$$

795 B.6 Proof of lemma 7

796 **Lemma 7.** For epoch s holds next inequality

$$\begin{aligned} f(w_{n/b+1}^{(s)}) &\leq f(w_0^{(s)}) - \alpha \theta^2 \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2 + \left(-\frac{1}{4\theta} + \delta + 2\alpha + (\theta + 2\alpha\theta^2) \frac{18e\delta^2 n^2}{b^2}\right) \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2 \\ &\quad + (\theta + 2\alpha\theta^2) \frac{16e\delta^2 n^2}{b^2} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2 \end{aligned}$$

797 *Proof.* Recall Lemma 3:

$$f(w_{t+1}^{(s)}) \leq f(w_t^{(s)}) + \left(-\frac{1}{4\theta} + \delta\right) \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 + \theta \|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2.$$

798 Multiply Lemma 6 by $\alpha > 0$ and add to the above:

$$f(w_{t+1}^{(s)}) \leq f(w_t^{(s)}) - \alpha\theta^2 \|G_\theta(w_t^{(s)})\|^2 + \left(-\frac{1}{4\theta} + \delta + 2\alpha\right) \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 \\ + (\theta + 2\alpha\theta^2) \|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2.$$

799 Summing over $t = 0, \dots, n/b - 1$:

$$f(w_{n/b}^{(s)}) \leq f(w_0^{(s)}) - \alpha\theta^2 \sum_{t=0}^{n/b-1} \|G_\theta(w_t^{(s)})\|^2 + \left(-\frac{1}{4\theta} + \delta + 2\alpha\right) \sum_{t=0}^{n/b-1} \|w_{t+1}^{(s)} - w_t^{(s)}\|^2 \\ + (\theta + 2\alpha\theta^2) \sum_{t=0}^{n/b-1} \|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2.$$

800 Substituting Lemma 1 into each $\|v_t^{(s)} - \nabla(f-f_1)(w_t^{(s)})\|^2$, then Lemma 2 into the initial error term,
801 and bounding the double sum by $\sum_{t=1}^{n/b-1} \sum_{j=0}^{t-1} \|j\|^2 \leq \frac{n}{b} \sum_j \|j\|^2$:

$$f(w_{n/b}^{(s)}) \leq f(w_0^{(s)}) - \alpha\theta^2 \sum_{t=0}^{n/b-1} \|G_\theta(w_t^{(s)})\|^2 \\ + \left(-\frac{1}{4\theta} + \delta + 2\alpha + (\theta + 2\alpha\theta^2) \frac{18e\delta^2 n^2}{b^2}\right) \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2 \\ + (\theta + 2\alpha\theta^2) \frac{16e\delta^2 n^2}{b^2} \sum_{j=0}^{n/b-1} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2. \quad \square$$

802 B.7 Proof of Theorem 1

803 *Proof.* Starting from Lemma 7, we set the inner-epoch coefficient equal to minus the inter-epoch
804 coefficient, so that the two sums telescope across epochs. This requires:

$$\frac{1}{4\theta} - \delta - 2\alpha - (\theta + 2\alpha\theta^2) \frac{18e\delta^2 n^2}{b^2} = (\theta + 2\alpha\theta^2) \frac{16e\delta^2 n^2}{b^2}.$$

805 Introduce dimensionless variables $B = \theta/\delta > 0$, $A = \alpha/\delta > 0$. Then:

$$\frac{1}{4B} - 1 - 2A - (B + 2AB^2) \frac{18en^2}{b^2} = (B + 2AB^2) \frac{16en^2}{b^2}. \\ \frac{1}{4B} - 1 - 2A - \frac{18en^2}{b^2} B - \frac{36en^2}{b^2} AB^2 = \frac{16en^2}{b^2} B + \frac{32en^2}{b^2} AB^2.$$

806 Gathering all terms with A on the right:

$$\frac{1}{4B} - 1 - \frac{34en^2}{b^2} B = \left(2 + \frac{68en^2}{b^2} B^2\right) A.$$

807 Hence:

$$A(B) = \frac{\frac{1}{4B} - 1 - \frac{34en^2}{b^2} B}{2 + \frac{68en^2}{b^2} B^2} > 0 \tag{10}$$

808 Let

$$C := (\theta + 2\alpha\theta^2) \frac{16e\delta^2 n^2}{b^2} > 0.$$

809 Then

$$f(w_{n/b+1}^{(s)}) \leq f(w_0^{(s)}) - \alpha\theta^2 \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2 - C \sum_{j=0}^{n/b} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2 \\ + C \sum_{j=0}^{n/b} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2.$$

810 Since $w_0^{(s)} = w_{n/b+1}^{(s-1)}$, we can rearrange:

$$f(w_{n/b+1}^{(s)}) + C \sum_{j=0}^{n/b} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2 \leq f(w_{n/b+1}^{(s-1)}) + C \sum_{j=0}^{n/b} \|w_{j+1}^{(s-1)} - w_j^{(s-1)}\|^2 \\ - \alpha\theta^2 \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2.$$

811 Define the Lyapunov function:

$$\Phi^{(s)} := f(w_{n/b+1}^{(s)}) + C \sum_{j=0}^{n/b} \|w_{j+1}^{(s)} - w_j^{(s)}\|^2.$$

812 Then:

$$\Phi^{(s)} \leq \Phi^{(s-1)} - \alpha\theta^2 \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2.$$

813 Rearranging and dividing by $n/b + 1$:

$$\frac{1}{n/b + 1} \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2 \leq \frac{\Phi^{(s-1)} - \Phi^{(s)}}{\alpha\theta^2(n/b + 1)} \leq \frac{\Phi^{(s-1)} - \Phi^{(s)}}{\alpha\theta^2 \frac{n}{b}}.$$

814 Averaging over epochs $s = 1, \dots, S$:

$$\frac{1}{(n/b + 1)S} \sum_{s=1}^S \sum_{t=0}^{n/b} \|G_\theta(w_t^{(s)})\|^2 \leq \frac{\Phi^{(0)} - \Phi^{(S)}}{\alpha\theta^2 \frac{n}{b} S} \leq \varepsilon^2.$$

815 Decompose $\Phi^{(0)} - \Phi^{(S)} = \Delta_1 + C \Delta_2$, where

$$\Delta_1 := f(w_{n/b+1}^{(0)}) - f(w_{n/b+1}^{(S)}), \quad \Delta_2 := \sum_{j=0}^{n/b} \|w_{j+1}^{(0)} - w_j^{(0)}\|^2 - \sum_{j=0}^{n/b} \|w_{j+1}^{(S)} - w_j^{(S)}\|^2.$$

816 Hence:

$$\frac{\Delta_1}{\alpha\theta^2 \frac{n}{b} S} + \frac{C \Delta_2}{\alpha\theta^2 \frac{n}{b} S} \leq \varepsilon^2.$$

817 Required number of epochs S :

$$S \sim \frac{\Delta_1}{\alpha\theta^2 \frac{n}{b} \varepsilon^2} + \frac{C \Delta_2}{\alpha\theta^2 \frac{n}{b} \varepsilon^2}.$$

818 Since each epoch uses n/b steps of batch size b (i.e., n oracle calls per epoch), the total oracle

819 complexity is $N_{oracle} = S \cdot n$:

$$N_{oracle} \sim \frac{\Delta_1}{\alpha\theta^2 \frac{1}{b} \varepsilon^2} + \frac{C \Delta_2}{\alpha\theta^2 \frac{1}{b} \varepsilon^2}.$$

820 Substituting $\alpha\theta^2 = \frac{AB^2}{\delta}$ and $C = \frac{16\mu B\delta(5/2-2B)}{2+68\mu B^2}$, where $\mu := \frac{en^2}{b^2}$ and $A = \frac{1/4-B-34\mu B^2}{B(2+68\mu B^2)}$, we
821 obtain:

$$N_{\text{oracle}} = \frac{b\delta}{B(1/4 - B - 34\mu B^2)\varepsilon^2} \left[(2 + 68\mu B^2) \Delta_1 + 16\mu B \delta \left(\frac{5}{2} - 2B \right) \Delta_2 \right].$$

822 Since $1/4 - B - 34\mu B^2 > 0 = 1/4 - B - 34en^2/b^2 B^2$ requires $B = O(b/n)$, we search for the
823 step size in the form $B = cb/n$ with $c > 0$. The optimal c^* satisfies

$$c^* = \sqrt{\frac{-13 + \sqrt{185}}{272e}} \approx 0.028.$$

824 Then, the N_{oracle} in terms of $\mathcal{O}(\cdot)$:

$$N_{\text{oracle}} = O\left(\frac{n\delta\Delta_1}{\varepsilon^2}\right) + O\left(\frac{n^2\delta^2\Delta_2}{b\varepsilon^2}\right). \quad (11)$$

825 For $b = n^\beta$, the oracle complexity becomes:

$$N_{\text{oracle}} = O\left(\frac{n\delta\Delta_1}{\varepsilon^2}\right) + O\left(\frac{n^{2-\beta}\delta^2\Delta_2}{\varepsilon^2}\right).$$

826

□

827 **C Technical appendices and supplementary material**

828 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
829 paper submission before the full submission deadline (see above). You can upload a ZIP file for
830 videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the
831 technical appendices.

832 Note: Think of the appendix as “optional reading” for reviewers. The paper must be able to stand
833 alone without the appendix; for example, adding critical experiments that support the main claims to
834 an appendix is inappropriate.